

Models of human perception

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As communications systems have increased in complexity it has become increasingly difficult to measure their performance objectively. In particular, when signals are compressed for more efficient transmission, conventional engineering metrics fail to predict the performance experienced by the end user — the customer. A new generation of objective measurement techniques is emerging based on models of human perception. Such models are potentially able to predict subjective performance for a wide range of transmission technologies. This paper introduces auditory and visual perceptual models and describes the development at BT Laboratories of models suitable for performance assessment. Early applications of the models are also described.

1. Introduction

The quality of a communications system is that perceived by the customer. There are many applications for objective performance metrics that relate directly to perceived quality. In a competitive market the ability to measure quality could become a key market differentiator.

Modern communications links frequently include compression to reduce the bandwidth required to transmit audio or video. Other nonlinear processes, such as echo cancellation, are also becoming increasingly common. The complexity of these systems makes them unsuitable for analysis using conventional signal processing techniques. End-to-end assessment of network quality must be based on what the customer would have heard or seen.

The main benchmark of customer opinion is the subjective test [1, 2], which is able to measure perceived quality in controlled experiments but is impractical for use in monitoring a network. This paper discusses computer models of human perception that can predict subjective test results and are suitable for large-scale applications.

Auditory perceptual models such as those of Hollier [3] and Beerends [4] are now being used for measuring telephone network quality and are described first. Visual perceptual models are at an earlier stage of development; section 3 presents an overview of one under construction at BT Laboratories. The discussion closes with the future direction of research in these fields and example applications of perceptual models, including non-intrusive and conversational techniques. A further paper [5] describes multi-modal perceptual models to take account of the

interactions in quality perception between the different senses.

These techniques do not attempt to model the complete extent of human perception — a goal that seems almost impossible to achieve. They aim to model properties of the human senses and behaviour that determine perception of the quality of a communications system. The success of BT's perceptual model for telephone speech quality assessment indicates that this is a useful way to proceed.

2. Auditory models

This section examines the important large-scale properties of hearing, how they can be modelled, and how auditory models are used in audio coding and quality assessment. Current research on BT's auditory model is discussed. Applications of auditory models are described in more detail in section 5.

2.1 Key properties of hearing

Two recordings that are indistinguishable to a human listener may have very different waveforms. A more perceptually relevant representation is therefore required if the signals are to be compared appropriately. Modelling the dominant properties of the auditory system — threshold, loudness response and masking — enables this to be achieved.

These properties can be understood, at least in qualitative terms, in the context of the physiology of the auditory system. Sounds are gathered by the outer ear and transmitted through the ear canal and the middle ear — an

intricate mechanical system that matches the impedance of the air-filled outer ear and the fluid-filled inner ear. The inner ear is a long tapered tube, through which a membrane (known as the basilar membrane) passes. Along this membrane are hair cells, which appear both to provide amplification and convert the acoustic waves to electrical impulses in the auditory nerves; different points along the basilar membrane respond to different frequencies. The dependence of firing rate with intensity accounts for threshold and loudness perception. Masking is caused by neural firings locking on to the loudest signals, suppressing quieter signal components, and by fatigue from loud sounds of long duration.

An excellent introduction to the properties of hearing is given by Moore [6]. Many models have been constructed by psychologists to investigate and explain characteristics of hearing; one of the most popular is the auditory image model (AIM) [7].

2.2 Auditory models in audio coding

A practical auditory model can be constructed with certain assumptions about listening equipment, signal levels, and knowledge of the properties of a listener with average hearing. This can then be applied to compressive coding, where the model is used to determine changes that can be made with minimal impact on perceived quality. Conversely, in audio quality assessment, the model's output is used to make the comparison between an original and a degraded signal to quantify audible errors introduced by the system.

High quality audio requires a large amount of data, making it expensive to transmit or store. An auditory model can enable the data to be compressed, bringing significant cost savings and making new applications possible. This is achieved by discarding inaudible parts of the signals, or making approximations which make as little audible difference as possible.

The MPEG family of audio coders [8] accomplish this with a model of masking. To illustrate this Fig 1 gives a simplified structure of the MPEG coder used to convert an audio signal to a compressed digital representation. The receiving end includes a decoder which resynthesises the

signal from the coded data; it does not itself include a perceptual model. The term *codec* is given to the combination of a coder and the corresponding decoder.

In this application, the perceptual model estimates the ratio of the signal level to the masking level at a given frequency. This gives the amount of noise that will be masked at each frequency, and is used to select the quantisation of the signal that provides the most compact digital representation with minimal audible difference.

Thus the compression process aims to maximise the quality of the coded signal, within the constraints imposed by the coder and model structure. The results are impressive — the MPEG-4 advanced audio coder promises to deliver compact disk (CD) quality audio at compression rates of 24:1, although this currently requires a large amount of computing power. To put this in context, this could extend the capacity of a CD to hold about 30 hours of audio, or allow the transmission of live broadcast-quality stereo audio over a single ISDN telephone line.

Speech coders have typically modelled the speech generation mechanism rather than hearing. Now that computationally tractable perceptual models are becoming available, it is likely that they will be used to enable higher quality or even greater bit-rate reductions. The main difference between a perceptual speech coder and the simplified audio coder described above is the inclusion of an iterative analysis/synthesis stage.

The distortions introduced by one type of compressive coder will be very different from another; unlike classical linear systems, the relationship between input and output is nonlinear, time-varying and signal-dependent. Subjective testing has, until recently, been the only way to compare coders for validation and standards selection. A subjective test consists of material — in this case, audio recordings — which is degraded in different conditions. A panel of subjects hear recordings passed through all conditions and vote on each. The votes are averaged to produce a mean opinion score (MOS) that gives a measure of the quality of each condition. The most common opinion scale in telephony applications is listener quality, a 5-point scale from 'excellent' to 'bad' [1].

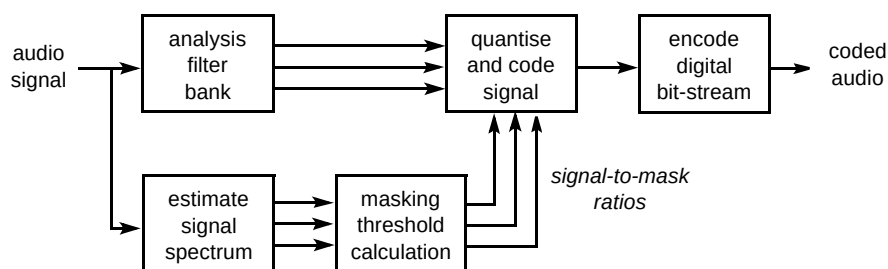


Fig 1 Simplified MPEG audio coder structure.

2.3 Auditory models in quality assessment

This section discusses the key stages in an auditory model and shows how they may be applied to objective quality assessment. BT's telephone speech quality assessment model is known as PAMS, the perceptual analysis/measurement system. PAMS makes no assumptions about the linearity of the system it is measuring, comparing instead the auditory representations of the original (reference) signal and the degraded output of the system.

Figure 2 shows the structure of the auditory transformation used in PAMS and described in Hollier [3]. The first stage models the filtering that takes place in the telephone handset, between the handset and the ear, and through the outer and middle ear. A bank of linear filters represents the frequency selectivity of the basilar membrane. The neural response is simulated by smoothing the power of the outputs of these filters and scaling according to results of equal loudness experiments. The result is a surface that represents the intensity of sound perceived by the auditory system at a given time and frequency band. An example plot for the sentence 'He paused and finished his drink' is shown in Fig 3.

The time-frequency representation provided by the auditory transform is useful as it reduces redundancy in the signal. This makes it possible to perform a representative comparison between signals without making any assumptions (such as linearity) about the transmission network.

This transform is based loosely on that of Wang [9], with an auditory filterbank similar to that of Sekey [10] and phon equal loudness contours as defined in ISO Standard 226 [11]. A key innovation in PAMS was the separation of the perceptual model into three sections — auditory transformation, extraction of parameters describing the amount and distribution of degradations, and mapping to subjective opinion. The model was developed by Hollier and is outlined elsewhere [3]. Recent refinements have focused on making the model's quality predictions robust with real telephone networks.

The structure of PAMS is illustrated in Fig 4. The model compares speech at the input and output of the system under test — a telephone network connection or codec. This configuration is known as intrusive assessment as the input signal is specified. Audible errors are estimated from the difference of the auditory transforms of the reference and degraded signals (the 'error surface').

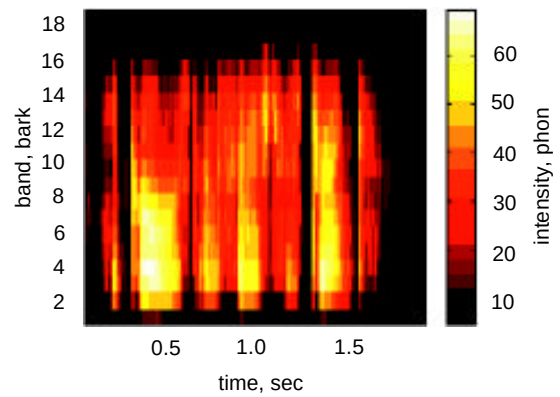


Fig 3 Auditory sensation surface.

Parameters describing the amount and distribution of degradations are extracted from this. For example, the average of positive errors gives a measure of the amount of noise that has been added to the degraded signal. A non-linear function is used to map between the error parameters and perceived quality as measured in subjective tests. This function is trained across a large number of subjective listening tests and is different for each opinion scale. PAMS predicts both listening quality and listening effort as defined in the ITU P.800 standard [1].

The emphasis in the development of PAMS is that of robustness for application on real networks. This distinguishes it from other models, notably the ITU-T P.861 standard and its recent annex, which are optimised for the assessment of codecs and are not designed to take account of large-scale behaviour of telephone networks [12, 13].

2.4 Artificial speech

An objective quality assessment model such as PAMS requires a reference signal to be injected into the network, for comparison with the network's output. The problem with using recorded natural speech is that it contains a good deal of repetition; tests would need to be of many minutes in duration to be comprehensive.

An artificial speech test signal has been developed at BT Laboratories (BTL) for use with both one-way and conversational quality assessment models. The most frequently occurring speech sounds (phonemes) from each phonetic group, as well as their first-order transitions, are identified by analysing conversational transcriptions. They are represented by a relatively small number of utterances, which are passed through BT's state of the art text-to-

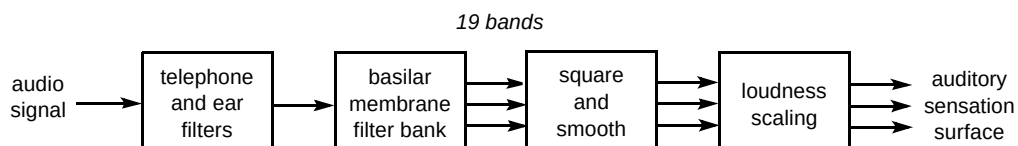


Fig 2 Structure of auditory transform.

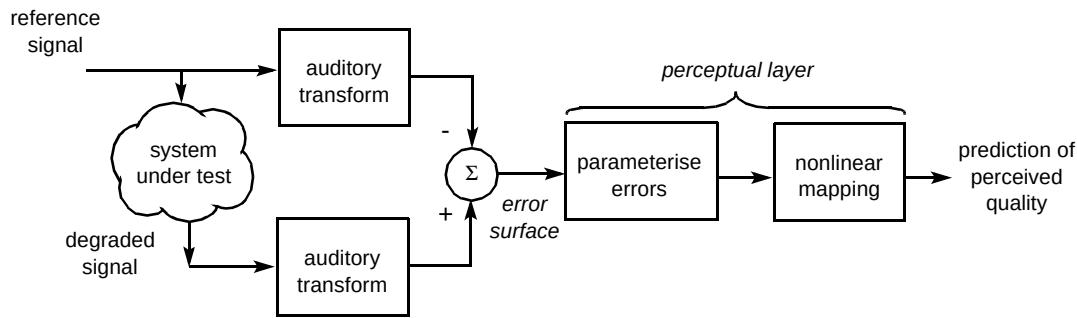


Fig 4 Structure of BT's PAMS perceptual model.

speech system, Laureate [14], to generate the test signal. British and American English versions have already been developed, taking about 32 seconds to include both male and female talkers. A measurement can therefore fully exercise the network while avoiding the redundancy of normal spoken language. This enables representative tests in relatively short periods of time, with a very significant improvement over existing standards.

2.5 Current research

The development of auditory models at BTL is continuing. The aims are to enhance the accuracy and robustness of PAMS, and extend it to wideband audio. A characteristic of recent packet-based networks (such as voice over Internet protocol) is variable delay. This defeats most speech quality assessment algorithms — changes in delay cause the models to compare different sections of speech, usually resulting in a quality score that is much too low. PAMS has recently been updated to correct this problem.

Work in the ITU-R has led to the drafting of a standard for the assessment of audio codecs such as MPEG. In contrast to telephony, which transmits signals from 300—3400 Hz, these are often wideband (20—20 000 Hz) and in stereo. Ways are being examined in which PAMS may be extended to assess audio systems from end to end, potentially using the same equipment as for telephony. A wideband version of PAMS would be suitable for inclusion in the multi-modal assessment framework under development at BTL [3, 5].

3. Visual models

As with hearing, vision has physiological properties that make some features present in visual stimuli very difficult or impossible to perceive. Compression algorithms, such as JPEG or MPEG, rely on these properties to reduce the amount of information to transmit in video streams. The quality of the resulting still or moving images cannot therefore be evaluated by simple comparison of the original and degraded signals. Properties of human vision have to be included in the assessment of perceived quality.

3.1 Overview

Most of the different approaches to visual modelling that have been reported are specialised to a particular application or video distortion. For instance, many models are designed to detect fine degradation which might occur in broadcast quality sequences. Karunasekera proposed a model [15] which is especially designed to detect blockiness.

At BTL, a visual perceptual model is being developed. It is meant to be as general as possible in order to cover a wide range of applications, from high-definition television to low-end videoconferencing, and even virtual reality. To meet this requirement, a more complex architecture has to be used. The model detects artefacts (particularly errors) which are physically perceptible to a human viewer, then determines how annoying any detectable errors are likely to be. The goal is to identify how well the main information content of the image has been preserved. In this way, it should be possible to determine if highly degraded sequences can still be understood by the user. This will enable assessment of even the lowest picture quality systems such as low bandwidth videoconferencing. The approach is also compatible with the assessment of coding algorithms using reconstruction, and potentially more abstract representations such as artificial head and shoulders.

The model proposed in this paper contains two stages. The first — the sensory layer — accounts for the physical sensitivity of the human visual system to given stimuli. The second — the perceptual layer — estimates the subjective annoyance caused by the remaining visible errors. This separation is analogous to that used in the auditory model described in section 2.3. The next two sections of the paper describe these stages of the model.

3.2 Sensory layer

The first part of the model, the sensory layer, emulates the physical properties of the human visual system. Its objective is to predict those details that are not physically perceptible, and to emphasise those to which the visual

system is more sensitive. To do so, it emulates the two main characteristics of the visual system which have an influence on the physical perceptibility of an artefact:

- spatio-temporal sensitivity of the eye (i.e. the human visual filter),
- spatio-temporal masking.

Figure 5 gives a representation of the sensory layer, showing how these two properties are emulated.

Firstly, the original and degraded sequences are filtered in temporal and spatial frequency by the human visual filter. This has the effect of weighting the information that they contain according to visual acuity. An error image is produced by subtracting the filtered original sequence from the degraded sequence. Both original and error sequences are then processed by the bank of filters, to split their content by spatial frequency and orientation. Typically, sixteen channels are used, comprising four orientations and four spatial frequencies. The energy content of each channel from the original sequence is used to compute the amount of masking to apply to each channel of the error image. Once this operation is complete, the error channels are added to reconstruct the masked error sequence.

Human visual filter

The human visual system is more sensitive to some spatial and temporal frequencies than others. Everyday experience teaches us that we cannot see details under a certain size. This effect is shown by Fig 6, where a grating of increasing spatial frequencies can be seen at different values of threshold according to the viewing distance. Spatial resolution is referred to in terms of spatial frequency, which is defined by counting the number of cycles of a sinusoidal pattern present per degree subtended at the eye. Closely spaced lines (fine details) correspond to high spatial frequencies, while large patterns correspond to low spatial frequencies. Once this concept is introduced, human vision can be compared to a filter, with peak (mid-range) sensitivity to spatial frequencies of around 8 cycles/degree and

insensitivity to high frequencies (60 cycles/deg and beyond). A similar filter characteristic can be applied in the temporal domain, where the eye fails to perceive flickering higher than 50 Hz (such as a neon light). The overall filter characteristic for both spatial and temporal frequency can be represented as a surface, as shown in Fig 7.

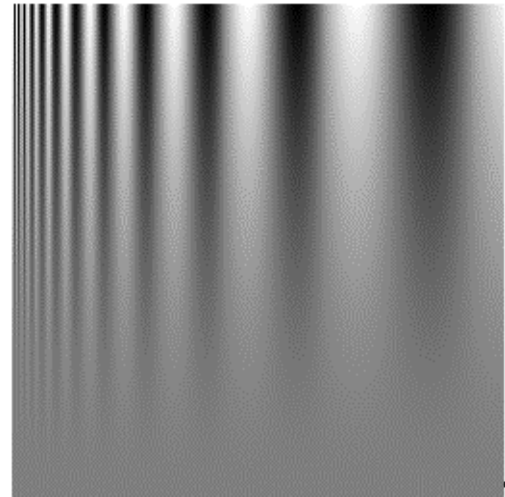


Fig 6 Sensitivity to different spatial frequencies.

The model of the human visual filter needs to be calibrated carefully since it precedes all subsequent model stages. This requires knowledge of the sensitivity of the eyes to each combination of spatial and temporal frequency, at each orientation. To obtain these results at the required precision involves a large number of subjective tests. This characteristic of the human visual system has been extensively studied in the past, and subjective tests were conducted to measure this transfer function. The results, shown in Fig 7, give the relative magnitude of visual sensitivity against spatial and temporal frequency.

This filter forms the first stage of a perceptual model for assessing image degradation. Details that are not visible can be removed, and therefore not counted as errors, while the perceptibility of other details is increased by the greater sensitivity at certain frequencies.

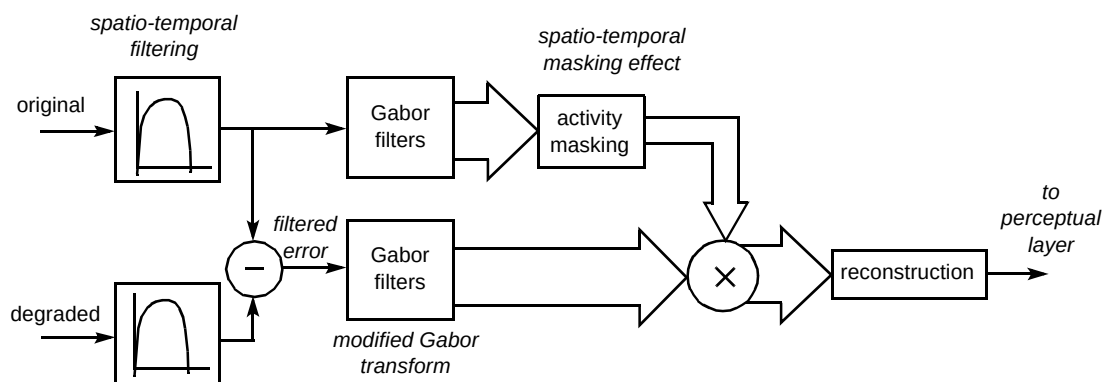


Fig 5 Sensory layer of the visual model.

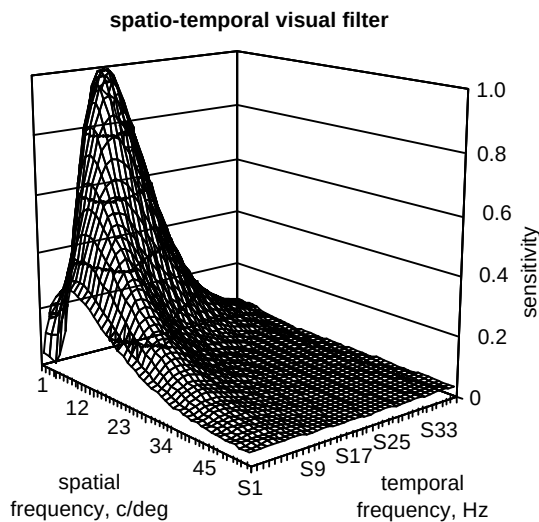


Fig 7 Human visual filter in frequency and temporal domain.

Masking effect of activity

The second aspect of vision modelled by the sensory layer is activity masking. On the JPEG reconstructed image shown in Fig 8, errors are present all over the picture. However, errors are less perceptible in some areas, such as the trees, because of the masking effect of the greater spatial activity present in this area. The temporal counterpart of this effect decreases the visibility of details as the rate of movement increases.

The origins of activity masking can be understood by considering the organisation of the primary cortex, the first stage of the brain responsible for visual processing. Each part of the cortex is sensitive to a certain region of the retina. The incoming image stream is divided into groupings (known as channels) of spatial frequency, temporal frequency and orientation. The 'next stage' of the brain will process the image stream as a set of channels, each accounting for a combination of spatial/temporal frequency and orientation in the corresponding area of the retina. Once a given channel is excited, it tends to inhibit its neighbours, making it more difficult to detect other details that are close in proximity, spatial or temporal frequency, and orientation.

For the sensory layer to simulate activity masking, it is necessary to know the amount of inhibition a channel causes to its neighbours. This information is obtained by studying the masking produced by representative samples of channels, in terms of spatial/temporal frequency and orientation characteristics.

The masking effect is modelled by calculating the amount of activity present in each combination of spatial frequency and orientation within an image. This calculation can be performed using a Gabor function, a flexible form of bandpass filter. However, it is desirable to be able to reconstruct the image from the channels once the masking effect has been calculated and applied to the error image. This cannot be performed with a bank of Gabor filters, since they are not invertible. Other functions can be used, such as the one described by Simoncelli [16]. An invertible modified Gabor function is used in this application and was proposed by Van Claster [17] and Van der Plancke [18].

The frequencies used in the filter bank are chosen as a harmonic series from the highest frequency present in the image. This ensures that the response of the whole filter bank is flat, and the image can be reconstructed by combining all the channels.

Calibration of this model of masking requires data describing how spatial/temporal frequency of a given orientation decreases the visibility of another stimulus. This information cannot be obtained as a complete description as the number of combinations is very large. Therefore, the separate influence of each parameter is measured. First the masking effect of a background on a stimulus is measured according to the relative orientation between the two. Then the effect of spatial and temporal frequency difference between masker and stimulus is measured. Finally, the two characteristics are combined by interpolating between common measured points.

3.3 Perceptual layer

Having modelled any errors present in an image sequence, it is next necessary to predict their subjective significance. If a simple comparison between original and degraded frames is used then certain types of error, such as



Fig 8 Original and JPEG reconstructed image.

delay in the image or a horizontal/vertical shift, result in large amounts of error all over the frame, but would not be noticeable to a user. This problem is usually addressed by employing frame realignment [19]. However this simple method does not fully account for the subjectivity of the error, since it does not allow for other common defects such as degradation of elements in the compressed sequence.

The disappearance of an edge is usually noticeable and annoying to the user, unless it is perceived as a part of a texture. Similarly, texture disappearance is not visible as part of a background, but becomes highly visible if the texture defines a shape, as in Fig 9. These factors are addressed by the perceptual layer, which gives an indication of error subjectivity by determining an approximation to the semantics of the image.

Most existing visual models comprise only a sensory layer, the perceptual layer typically being elementary [15, 20]. For example, the assumption that large errors will dominate subjectivity can form the basis of a simple perceptual layer [20] where it is assumed the subject has insufficient time to view the whole image. Other approaches are also used, such as a model of the temporal aggregation of errors [21]. However, none of these approaches addresses the relative importance of all errors present in the image.

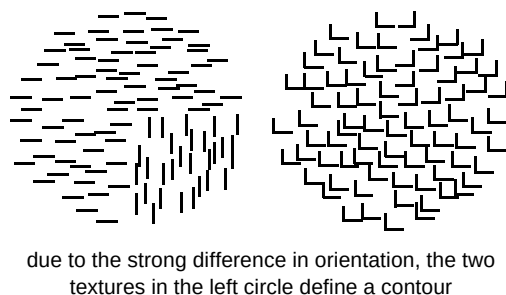


Fig 9 Shape defined by texture.

The basis for the proposed perceptual layer is that elements present in the image are not of equal semantic importance. An error will be more perceptible if it coincides with a feature contributing to the semantics of the image. For example, a distortion present on one of the edges forming part of a texture will be less perceptible than the same error on an independent edge. This is because an edge forming part of a texture carries less information than an independent edge [22]. If, however, a textured area defines a contour as in Fig 9, an error that changes the properties of the texture can be as important as an error on an independent edge.

Most authors working in the field of pattern recognition consider either contours or boundaries to be the fundamental information carrier in image sequences [23]. These boundaries are usually defined as edges, but other classes can be included, such as textures, colours, movement or disparities between the eyes.

The first stage of the perceptual layer splits the incoming sequence into these perceptual components, as shown in Fig 10. Once the contour or boundaries are predicted for both original and degraded sequences, the perceptual layer can match them, identifying differences between them. The error image produced by established models (filtered and masked noise) provides an indication of the visible degradation of the image. However, the perceptual layer introduced here includes an estimate of the extent to which the semantic content is maintained and offers an improved prediction of the image intelligibility.

This perceptual layer is suitable for a greater range of video quality assessment applications, where prediction and comparison of the perceptual contours is necessary. A good example is given by very low bandwidth systems where a face is algorithmically reconstructed — which would be impossible for many of the visual models described in the literature to assess appropriately. The approach of comparing perceptual contours will also enable the assessment of synthetic/abstract representation of the images such as an animated talking head.

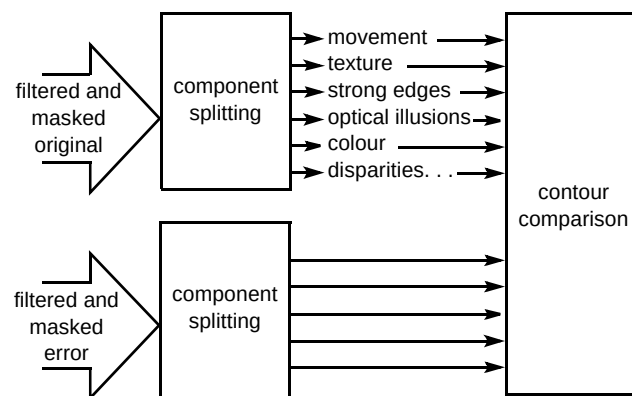


Fig 10 Overview of the perceptual layer.

3.4 Visual model performance

It is immediately problematic to try to locate information from an image by mathematical processing of pixel values. The pixel intensity level becomes meaningful only once processed by our visual knowledge of objects and shapes. Since reproducing this property of the human brain is out of the question (at least in the first version of the model), mathematical solutions have had to be relied upon to extract information which resembles that used by the brain as closely as possible.

On their own, extraction algorithms typically perform poorly. Each tends to create false contours, and fails to detect others. However, the combination of different algorithms increases the quality of the result, a visual object being often defined by many perceptual boundaries [23]. The next example shows the decomposition of an image by different means, such as edges, colours and textures.

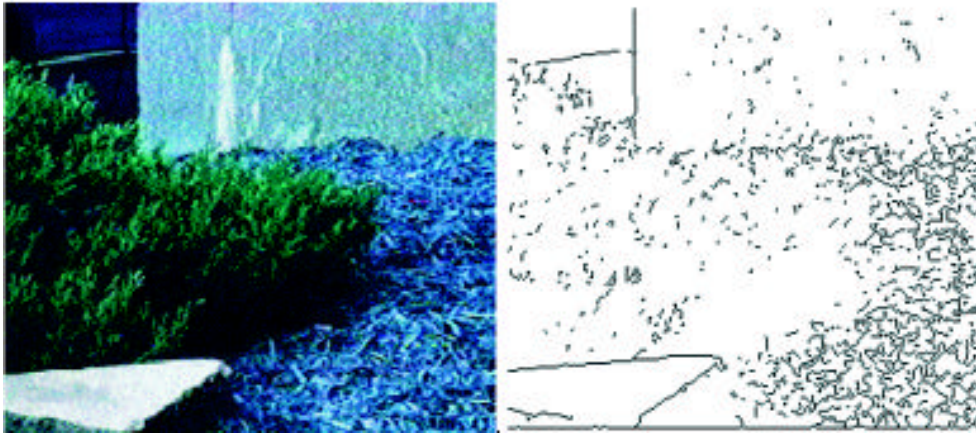


Fig 11 Result of SUSAN edge-extraction algorithm.

Figure 11 shows the original image and the result of analysis with an edge extractor (SUSAN [24]). As expected, the result is good at detecting the boundaries of large continuous surfaces, but failed to give meaningful information for the textured area. In this configuration, the detector indicates a pseudo-random pattern of small edges. Since texture is perceptually detected as a whole, and not a pattern of edges, this information is perceptually irrelevant.

The second boundary extractor used is the texture classifier. To perform this operation, the result given by the modified Gabor filter bank is used. Each channel gives the energy content at a given spatial frequency and orientation. This information is then processed by a K-mean algorithm, which will cluster the image into K categories. Figure 12 shows the result on the same image.

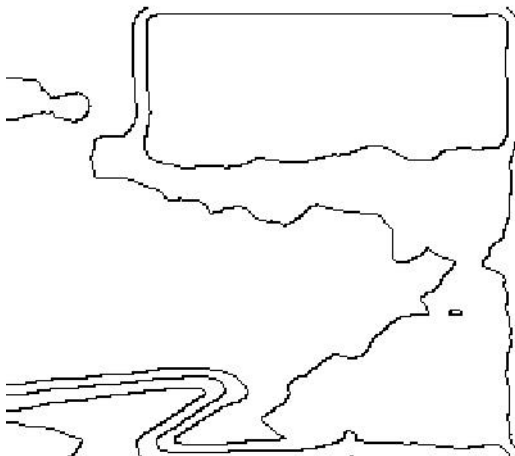


Fig 12 Texture segmented image.

This method separates the different regions of the image according to their texture content. This segmentation is close to the one we would obtain mentally — the area covered with plants is segmented into two, the grass and the bushes. From this point of view, the segmentation matches

with the perceptual information. Elsewhere in the image, the detector seems unable to distinguish between the car and the bushes.

The main problem of K-mean segmentation algorithms is the choice of the number of clusters, which determines the number of detected textures. This number varies according to the picture content and the efficiency of the algorithm to detect a given texture.

These first results remind us that extraction algorithms often fail to detect the relevant information present in the image. However, combination of their results will allow us to collect most of the boundaries present in the analysed sequences.

In summary, the visual model under development at BTL is constructed with two main stages — the sensory layer and the perceptual layer. The sensory layer reproduces the large-scale properties of vision, and has been calibrated with extensive subjective test data. The perceptual layer is based on novel image decomposition techniques that extract the semantically important boundaries within the image and then predict error subjectivity by comparing these boundaries. Initial testing of the perceptual layer demonstrates the potential of this approach.

4. Future research

The use of the auditory perceptual model with BT's networks has produced requirements for further development. The subjectivity of filtering speech and music, such as the process to limit the audio bandwidth of telephone networks, needs to be investigated to make PAMS capable of assessment at different bandwidths. Other areas of research include quality implications of background noise, the differences between audibility and annoyance, and the effect of degradations on different parts of speech.

The perceptual layer is the focus of current development in the visual model. This includes two main activities. The first is finding and improving techniques to extract perceptual contours such as textures and edges. In parallel, subjective tests must be carried out in order to calibrate the new model, by determining the relative importance of the different types of boundaries.

The quality of a multi-modal communications system depends strongly on interactions between the individual senses as well as effects such as synchronisation. Research into multi-modal perceptual modelling is investigating this behaviour. The goal is to construct a multi-modal perceptual model combining audio and visual models [3, 5].

The auditory and visual perceptual models seek a more general approach to quality assessment than is currently used in delivery technology such as audio and video codecs. As the models become mature there could be considerable benefit in including them in the network, assuring end-to-end quality.

5. Applications

The goal of this work has been to produce perceptual models that can take the inputs and outputs of a communications link and predict the quality that would have been perceived by the customer. Possible applications include extending the capability of subjective tests for evaluating or planning new communications networks, monitoring the quality of an installed network, and identifying problems so that quality can be maintained.

5.1 End-to-end assessment

There are many applications for the auditory perceptual model as an end-to-end audio quality measure. PAMS has already been used in BT to assist with commissioning and testing new networks. It is suitable for use in measuring and tracking the quality of an installed telephone network, for example, calling out an engineer if quality falls below a predetermined threshold.

As PAMS predicts the results of subjective listening tests with high accuracy, a new structure of subjective test may be considered. The objective auditory model can measure the quality of many more conditions than would be possible using a conventional test structure, with subjective opinion scores being used to verify the model's predictions.

Just like the auditory model, the visual model has many applications. Examples include continuous assessment of quality during the installation and configuration of a video system, the development of new coding software or hardware, and maintenance or quality assurance of an existing video link. As part of a transmission system, it could be used to optimise end-to-end quality against

compression ratio. Because the model covers many visual properties, the range of these applications is wide, from high-quality systems such as high-definition television or digital cinema, to lower quality services including videoconferencing or video over the Internet.

5.2 Conversational models

The perceptual model described in section 2.3 can assess the one-way quality of speech transmission through a network, taking no account of properties such as level, echo and delay. These affect the ease of conversing over the network even if unidirectional speech transmission is unimpaired. Conversational subjective tests are therefore the preferred method of subjective evaluation of whole telephone networks. Participants hold conversations over a number of different networks and vote on their perception of conversational quality.

Research at BTL has made significant steps to ensure that objective speech quality assessment is able to take account of these conversational factors. The aim is to produce a test methodology that is suitable for use in real networks and that performs representative assessments in a short time.

An auditory model forms part of BT's call clarity index (CCI), a technology for estimating conversational quality using only non-intrusive measurements across linear systems. This predicts performance based on measurements of echo, delay, level and noise, in combination with an auditory model to estimate the masking of speech by noise. The result is a single measure that can be used to describe the perceived quality of a telephone call [25]. CCI is already in use in BT's network.

Non-intrusive modelling is a key part of network quality measurement. For example, CCI could be used to monitor the quality of a large number of calls. A relatively small number of intrusive measurements would be made using PAMS to check the network's nonlinear properties, and could also be performed to investigate poor quality results reported in the non-intrusive measurements.

The same techniques have been used to weight PAMS scores to take account of these conversational impairments when measurements are made across nonlinear networks (such as those including speech codecs), which CCI does not itself take into account. Experiments indicate that this enables PAMS to be used successfully to predict the scores of conversational subjective tests.

Conversational assessment may be improved to take account of the relationship between transmission quality and the signals being carried. Most networks are designed to perform best when speech properties such as level are in some optimum range, and degrade with signals outside this

range. A key dependence is the level or loudness of speech, which varies naturally in conversation as the participants alter their behaviour because of noise in the environment and any impairment in intelligibility of the signals they hear. This can be modelled by an adaptation process (itself using a unidirectional model such as PAMS) which simulates level variations in speech. The resultant quality scores and levels, combined with echo and delay, give an estimate of conversational quality taking account of level dependence [26, 27].

Perhaps the ultimate goal of conversational modelling is to model the dynamic nature of a conversation while retaining the objectivity and repeatability of an automatic assessment system. The idea is for two devices, one at each end of a connection, holding a conversation with each other. (For an audioconference the same idea would apply with more devices.) In a recent patent application [28] a structure able to perform this dynamic conversational assessment was described. A simplified diagram of this is shown in Fig 13.

Each device has a conversational intent — the information it wishes to get across. This may be subdivided into parts of speech which are important to the conversation, and others which are of peripheral value. This information is exchanged and quality assessment performed continuously, enabling the network's behaviour to be tested when poor quality interrupts the conversation, both participants speak at the same time, or echo, noise or level changes impair understanding. The advantage of this approach over the methods described above is the ability to exercise the network using a representative conversational pattern that alters with delay and variations of quality with time.

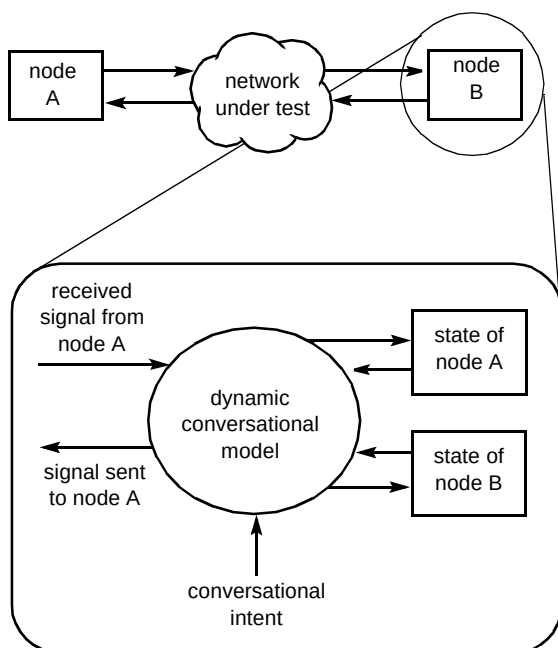


Fig 13 Changing conversational state.

6. Discussion

As the signal processing deployed in communications systems has become more complex, it has become necessary to develop a new generation of objective measures based on models of the human sense. Such models can predict the perceptibility of errors even when signals have been processed by highly nonlinear systems such as compressive codecs.

BT's auditory perceptual model is relatively mature. PAMS has been shown to be robust in assessing a wide variety of telephone networks and systems. The BT visual perceptual model is based on a novel technique comparing semantically important boundaries. This approach is tolerant of perceptually insignificant distortions such as small spatial displacements, and is intended to weight errors in a manner which is consistent with cognitive image interpretation. Future perceptual model developments have also been introduced, including new applications for auditory and visual models and their combination into a multi-modal model.

Acknowledgements

The authors would like to thank their colleagues at BT Laboratories who have collaborated in this work and assisted in the preparation of this paper.

References

- 1 ITU-T Recommendation P.800: 'Methods for subjective determination of transmission quality', (this Recommendation was previously numbered P.80) (1996).
- 2 ITU-T Recommendation P.911: 'Subjective audiovisual quality assessment methods for multimedia applications', (1998).
- 3 Hollier M P and Cosier G: 'Assessing human perception', *BT Technol J*, **14**, No 1, pp 206—216 (January 1996).
- 4 Beerends J G and Stemerdink J: 'A perceptual audio quality measure based on a psychoacoustic sound representation', *J Audio Eng Soc*, **40**, No 12, pp 963—974 (1992).
- 5 Hollier M P, Rimell A, Hands D and Voelcker R M: 'Multi-modal perception', *BT Technol J*, **17**, No 1, pp 35—46 (January 1999).
- 6 Moore B C J: 'An introduction to the psychology of hearing', 4th edition, Academic Press (1997).
- 7 Patterson R D, Robinson K, Holdsworth J, McKeown D, Zhang C and Allerhand M H: 'Complex sounds and auditory images', in Cazals Y, Demany L and Horner K (Eds): 'Auditory Physiology and Perception', pp 429—446, Pergamon Press (1992).
- 8 Brandenburg K and Bosi M: 'Overview of MPEG audio: current and future standards for low-bit-rate audio coding', *J Audio Eng Soc*, **45**, Nos 1/2, pp 4—21 (1996).
- 9 Wang S, Sekey A and Gersho A: 'An objective measure for predicting subjective quality of speech coders', *IEEE J Sel Areas in Comms*, **10**, No 5, pp 819—829 (1992).
- 10 Sekey A and Hanson B: 'Improved 1-Bark bandwidth auditory filter', *J Acoust Soc Am*, **75**, No 6, pp 1902—1904 (1984).

- 11 'Acoustics — normal equal loudness contours', ISO Standard 226 (1987).
- 12 ITU-T Recommendation P.861: 'Objective quality measurement of telephone-band (300—3400 Hz) speech codecs', (1996).
- 13 Atkinson D J: 'Proposed Annex A to Recommendation P.861', ITU-T contribution COM12-24 (December 1997).
- 14 Page J H and Breen A P: 'The Laureate text-to-speech system — architecture and applications', BT Technol J, 14, No 1, pp 57—67 (January 1996).
- 15 Karunasekera A S and Kingsbury N G: 'A distortion measure for blocking artifacts in images based on human visual sensitivity', IEEE Transactions on Image Processing, 4, No 6, pp 713—724 (June 1995).
- 16 Simoncelli E P and Freeman W T: 'The steerable pyramid: a flexible architecture for multi-scale derivative computation', IEEE Second International Conference on Image Processing, Washington, 3, pp 444—447 (November 1995).
- 17 Van Cluster O: 'Perceptual post processing of the coding noise for digital pictures', Master's thesis, Université de Louvain la Neuve (June 1993).
- 18 van den Branden Lambrecht C J: 'Perceptual models and architectures for video coding applications', Thèse No 1520, EPFL, Lausanne (1996).
- 19 Webster A: 'Draft new recommendation on multimedia communication delay, synchronization, and frame rate measurement', ITU-T contribution COM 12-29 (December 1997).
- 20 Lukas X J and Budrikis Z L: 'Picture quality prediction based on a visual model', IEEE Transactions on Communications, COM-30, No 7, pp 1679—1692 (July 1982).
- 21 Tan K T, Ghanbari M and Pearson D E: 'A video distortion meter', Informationstechnische Gesellschaft, Picture Coding Symposium, Berlin (September 1997).
- 22 Ran X and Favardin N: 'A perceptually motivated three-component image model — Part II: application to image compression', IEEE Transactions on Image Processing, 4, No 4, pp 713—724 (April 1995).
- 23 Scassellati B M: 'High-level perceptual contours from a variety of low-level physical features', Masters Thesis, Massachusetts Institute of Technology (May 1995) — <http://www.ai.mit.edu/people/scasz/thesis.ps.gz>
- 24 Smith S M and Brady J M: 'SUSAN — A new approach to low level image processing', Technical report TR95SMS1c, Oxford Centre for Functional Magnetic Resonance Imaging of the Brain (1995) — <http://www.fmrib.ox.ac.uk/~steve/susan/susan.ps.gz>
- 25 Broom S R, Coackley P and Sheppard P J: 'Getting the message loud and clear — quantifying call clarity', British Telecommunications Eng J, 17, pp 66—72 (April 1998).
- 26 Reynolds R J, Hollier M P, Rix A W and Sheppard P J: 'Measurement of signal quality', International patent application PCT/GB98/01306 (16 May 1997).

27 Reynolds R J and Hollier M P: 'Extending listening models into bi-directional conversation models for network assessment', ITU-T delayed contribution COM12-D036 (February 1998).

28 Rix A W, Hollier M P, Reynolds R and Sheppard P J: 'Testing communications systems', International patent application PCT/GB98/01305 (16 May 1997).



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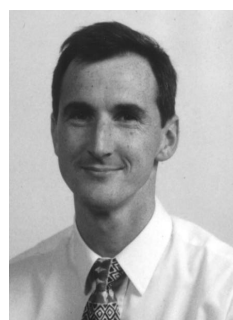
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